

Braillix print release - 2026-08-19

Verdict

Do not print the complete product set yet. The current production cam cannot install on the measured 7.5 mm motor shaft, and its ramp/linkage/spring behavior still needs the physical bench test below.

The checked PETG folder contains exactly eight files. They are either small fit tests, a mechanism test fixture, or the stack-independent mid plate. No enclosure, production cam, base plate, pod, or final top plate is released.

Tomorrow's print order

Required fit tests

- 1 motor_cam_socket_coupon.gcode - about 29 min / 2.62 g
- 2 base_interface_coupon.gcode - about 1 h 10 min / 7.97 g
- 3 pigtail_slot_coupon.gcode - about 42 min / 5.77 g
- 4 pogo_receiver_coupon.gcode - about 1 h 10 min / 12.79 g
- 5 top_interface_coupon.gcode - about 1 h 24 min / 10.70 g
- 6 hardware_fit_coupon.gcode - about 2 h 10 min / 20.03 g

Stop after any print with poor first-layer adhesion, separated walls, closed holes, heavy stringing, or visible warping. A bad coupon means the larger part is not safe.

Mechanism test fixture

- 1 cam_linkage_test_fixture.gcode - about 2 h 10 min / 18.80 g

This fixture is useful only with `cad/stl/cam_linkage_test_resin_set.stl`, one real 2 mm OD spring, and the procedure in `[MECHANISM_BENCH_TEST.md]`(`MECHANISM_BENCH_TEST.md`). The resin set contains the real cam, dot-5 linkage, dot-6 linkage, and dot insert. It is intentionally STL-only because the resin printer, resin, and service settings are not specified.

Stack-independent production part

- 1 mid_plate.gcode - about 1 h 16 min / 10.07 g

Print the mid plate only after the first coupon proves the PETG/profile is behaving normally. Its geometry already uses the measured 28.1 mm motor body, -7.5 mm body offset, and 34.7 mm ear spacing, and it survives the pending +4 mm upper-stack option.

Checked PETG contract

All eight checked G-code files were freshly sliced from the current STLs with:

- NuMaker PETG HS: 235 C first layer / 230 C normal
- bed 80 C
- layer height 0.16 mm
- 5 wall loops
- 40% infill
- 6 bottom + 6 top shell layers
- 8 mm brim
- top-surface ironing
- supports disabled
- no G2/G3 arc commands

`tools/validate_print_assets.py` verifies the exact checked manifest, embedded settings, Kobra Neo motion envelope, closed/manifold topology, expected component count, and committed STL equality with independent fresh OpenSCAD renders.

Coupon maps and acceptance

Motor/cam socket coupon

The one-, two-, and three-hole tabs are 3.1, 3.2, and 3.3 mm double-flat sockets.

Use the smallest socket that:

- slides fully over the unpowered motor shaft;
- reaches and sits flat on the measured 9 mm collar;
- does not split, whiten, or require hammering;
- can still be removed using the tab.

Do not heat, glue, hammer, or power the motor during this test.

Base-interface coupon

This has three disconnected pieces, left to right:

- dia 34 puck: real dia 29 x 1.5 mm underside motor seat around the dia 10 shaft opening. Reject it if the bridged locating roof sags enough that the motor cannot sit flat without rocking.
- 2 mm bar: M4 pilot holes 3.3 / 3.4 / 3.5 mm. This selects diameter/engagement only, not length.

Use the smallest that accepts an M4x5/x6 thread-forming candidate with controlled thread formation and no whitening, splitting, or stripped PETG. Never use M4x10; after Option A approval, measure the ear/point and keep the installed tip ≥ 0.5 mm below the cam.

- three dia 6 x 12 mm posts: M2.5 clearance bores 2.9 / 3.0 / 3.1 mm. Use the smallest free-sliding hole that does not split the slender post. These reproduce the proposed taller Option-A posts; testing them does not authorize the production stack change.

Pigtail cable-slot coupon

The clipped corner marks the 5 x 3 mm end. The middle slot is the current 6 x 4 mm CAD value; the other end is 7 x 5 mm. Pass only the loose red/black leads, never the barrel body. Select the smallest opening that accepts both leads without scraping, pinching, whitening, or cutting their insulation. Record the selected width x height before regenerating the pod lid.

Pogo receiver coupon

Left to right, the upright bridged slots are:

- 10.0 x 8.0 mm - current production CAD;
- 10.2 x 8.2 mm;
- 10.4 x 8.4 mm.

All reproduce the real 4 mm enclosure wall. Accept a slot only if its roof prints open without cutting structural material and the measured opening is usable. The end cap itself remains TPU 95A only. No TPU G-code is released until the exact spool is known.

Top-interface coupon

The large keyed piece tests the actual M2.5 x 25 screw.

- clipped corner = top-left;
- columns left to right = counterbore diameters 5.4 / 5.6 / 5.8 mm;
- rows top to bottom = effective depths 1.3 / 1.5 / 1.7 mm;
- bottom plain holes left to right = 3.0 / 3.2 / 3.4 mm shank clearance.

Select the smallest/shallowest station where the real screw drops through, the head seats flat, and a finger/straightedge does not catch it.

The separate notched strip tests the real resin dot insert:

- notched end is left;
- left to right total clearance = 0.2 / 0.4 / 0.6 mm;
- production CAD currently uses 0.2 mm total, only 0.1 mm per side.

Select the smallest pocket where the insert seats without forcing or cracking and can be removed before gluing. The held top plate must be updated/re-sliced if the selected station is not the current 5.6 x 1.5 mm screw seat and 0.2 mm insert clearance.

Hardware-fit coupon

- top-left row 3.1 / 3.3 / 3.5 / 3.7: heat-set insert bores;
- bottom-left row 3.1 / 3.2 / 3.3: shaft double-flat clearances;
- top-right row 8.2 / 8.4 / 8.6: 8 x 1 mm magnet pockets;
- bottom-right row 1.7 / 1.8 / 1.9: Hall-body pocket depths.

Do not melt an insert or glue a magnet into a production shell before choosing the coupon fit.

HOLD - do not manufacture as final parts

Part	Blocking reason
top_plate	G-code moved to printing/gcode_HOLD_fit_unproven; real screw head and resin insert must pass the top-interface coupon
braille_cam	accidental 3.5 mm blind socket cannot install on the 7.5 mm shaft; current 64-state ramps still need the real dot-5/dot-6 test
base_plate	stack/Hall height depends on the selected repair; motor-seat bridge, M4 pilot, and slender post fits must pass the base-interface coupon
outer_box	final height depends on the stack decision; pogo carrier is unmeasured/unretained and the ULN2003 has no positive board retainer
esp32_pod_shell	must match final cell height; header tails have no floor/wire exit, dock pads have no retainer/lead route, and exact contact/header hardware is not owned
esp32_pod_lid	inline-pigtail geometry is corrected, but matching shell height is pending
linkage	final release needs resin dot-5/dot-6 ramp, spring-return, binding, and torque evidence
dot_insert	final release needs real resin fit, dome sliding, spring insertion, and 50-cycle evidence
nav_cap	supported/angled resin only; not a support-free PETG part
pogo_end_cap	TPU 95A only; receiver size and barb retention require a real insertion/removal test

Old 3-wall G-code is quarantined under printing/gcode_HOLD_unvalidated. The old print_batch folder is historical and marked DO NOT USE.

Confirmed corrections already applied

- motor body offset changed from -8.0 to measured -7.5 mm;
- motor ear spacing changed from 35.0 to measured 34.7 mm;
- top counterbores now provide 1.5 mm below the recessed reading surface;
- outer-box pogo windows are symmetric and fully through;
- pod dock-wall antenna subtraction that left 0.5 mm skin was removed;
- guessed unsupported panel-jack cradle was removed;
- pod lid now uses the owned inline pigtail with a 6 x 4 mm cable slot and tie post;
- pogo end cap now has a hollow TPU tongue and real barb geometry;
- stale 3-wall outputs were quarantined;
- stack-independent motor, base, top, hardware, pogo, pigtail, and mechanism test assets were added.

Pending production-stack decision

Recommended non-destructive repair:

- keep the motor shaft intact;
- seat the cam hub on the measured 9 x 2 mm collar;
- raise the upper mechanism and matching enclosures by exactly 4 mm;
- rebuild the cam with a true 7.7 mm blind double-flat socket and central roof;
- raise the Hall sensor on an island to preserve the 0.4 mm magnet gap.

This coordinated change is not applied without AJ's explicit approval because it changes every upper datum. The alternative keeps the current 58 mm enclosure only by cutting and deburring the motor shaft to about 5 mm above the

mounting face.

After the chosen stack repair, regenerate every dependent STL, repeat the real mechanism bench test, verify Hall switching at the real gap, select coupon dimensions, and only then create the final production G-code manifest.